

Bayesian inversion with optimal transport maps and Polynomial Chaos surrogates - Application to mechanical inverse problems

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Introduction

- ▶ Inverse problems are often **ill-posed** in the Hadamard sense (existence, uniqueness, regularity) $\Rightarrow$ necessity to use **regularization** techniques
- ▶ In contrast to classical deterministic approaches, **Bayesian inference** offers a probabilistic viewpoint for inverse problems (Kaipio and Somersalo, 2005; Tarantola, 2005)
- ▶ The **posterior distribution** arising from Bayesian inference is usually estimated through MCMC (*Markov Chain Monte Carlo*) (Robert and Casella, 2004)
- ▶ However, classical MCMC algorithms suffer from several drawbacks: auto-correlation, lack of clear convergence criterion, large amount of model calls...

$\Rightarrow$ Use of an alternative computational framework for Bayesian inference

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Bayesian inverse problems

- ▶ Let $\mathcal{M} : \mathcal{D}_{\mathbf{X}} \subset \mathbb{R}^d \longrightarrow \mathcal{D}_{\mathbf{Y}} \subset \mathbb{R}^n$ be a **model**
- ▶ The input **parameters** of $\mathcal{M}$ are represented by a $\mathcal{D}_{\mathbf{X}}$ -valued random variable $\mathbf{X}$, which is assumed to follow a **prior distribution** with density π (i.e. $\mathbf{X} \sim \pi(\mathbf{x})$)
- ▶ One assumes a probabilistic model for ($\mathcal{D}_{\mathbf{Y}}$ -valued) **observables** $\mathbf{Y}$ conditionnally on $\mathbf{X}$:

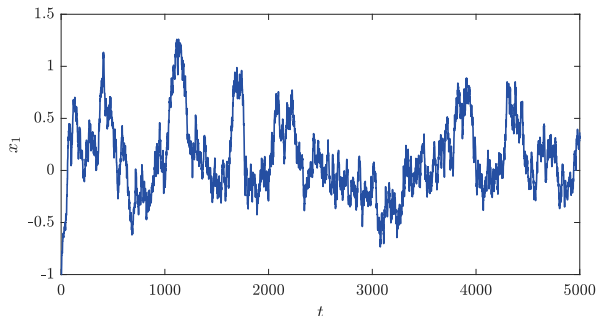
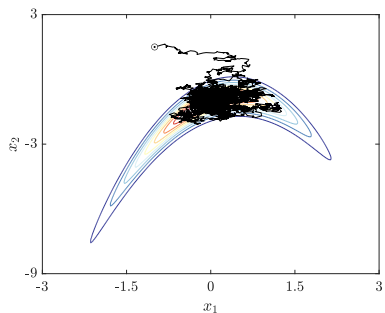
$$\mathbf{Y}|\mathbf{X} = \mathbf{x} \sim \pi(\mathbf{y}|\mathbf{x})$$

- ▶ Given $\mathbf{y} \in \mathcal{D}_{\mathbf{Y}}$, the density $\pi(\mathbf{y}|\mathbf{x})$ is seen as a function of $\mathbf{x} \in \mathcal{D}_{\mathbf{X}} \Rightarrow$ **likelihood** $\mathcal{L}_{\mathbf{y}}(\mathbf{x}) = \pi(\mathbf{y}|\mathbf{x})$
- ▶ **Bayes' theorem** enables to derive the **posterior density**:

$$\pi_{\mathbf{y}}(\mathbf{x}) = \pi(\mathbf{x}|\mathbf{y}) = \frac{\mathcal{L}_{\mathbf{y}}(\mathbf{x})\pi(\mathbf{x})}{C} \quad \text{where} \quad C = \int_{\mathcal{D}_{\mathbf{X}}} \mathcal{L}_{\mathbf{y}}(\mathbf{x})\pi(\mathbf{x})d\mathbf{x}$$

Bayesian computations with MCMC

- ▶ The **model evidence** C is rarely known analytically in practice $\Rightarrow$ only the **unnormalized** posterior $\tilde{\pi}_{\mathbf{y}}(\mathbf{x}) = \mathcal{L}(\mathbf{x})\pi(\mathbf{x})$ is known
- ▶ **MCMC** techniques: widely used to sample distributions whose densities are known only up to a normalizing constant $\Rightarrow$ sample auto-correlation, tedious tuning,... (Robert and Casella, 2004)



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Generalities on measure transport

- ▶ One defines prior and posterior probability **measures** ν and $\nu_{\mathbf{y}}$:

$$\nu(d\mathbf{x}) = \pi(\mathbf{x})d\mathbf{x} \quad \nu_{\mathbf{y}}(d\mathbf{x}) = \pi_{\mathbf{y}}(\mathbf{x})d\mathbf{x}$$

- ▶ Given a (measurable) map $T : \mathcal{D}_{\mathbf{X}} \longrightarrow \mathcal{D}_{\mathbf{X}}$, the **pushforward** $T_{\#}\nu$ of ν by T is defined by:

$$T_{\#}\nu(B) = \nu(T^{-1}(B))$$

where $T^{-1}(B) = \{\mathbf{x} \in \mathcal{D}_{\mathbf{X}} \mid T(\mathbf{x}) \in B\}$, for all Borel set $B \in \mathcal{B}(\mathcal{D}_{\mathbf{X}})$

- ▶ Likewise, the **pullback** $T^{\#}\nu_{\mathbf{y}}$ of $\nu_{\mathbf{y}}$ by T is defined by:

$$T^{\#}\nu_{\mathbf{y}}(B) = \nu_{\mathbf{y}}(T(B))$$

Generalities on measure transport

- ▶ The map T will be supposed to be sufficiently regular, *i.e.* at least a $\mathcal{C}^1$ -diffeomorphism
- ▶ In such a case, the pushforward and pullback measures $T_{\#}\nu$ and $T^{\#}\nu_{\mathbf{y}}$ admit densities:

$$T_{\#}\pi(\mathbf{x}) = \pi(T^{-1}(\mathbf{x})) |\nabla T^{-1}(\mathbf{x})|$$

$$T^{\#}\pi_{\mathbf{y}}(\mathbf{x}) = \pi_{\mathbf{y}}(T(\mathbf{x})) |\nabla T(\mathbf{x})|$$

- ▶ Important fact: if $\{\mathbf{x}^{(k)}\}_{1 \leq k \leq N} \stackrel{\text{i.i.d.}}{\sim} \pi(\mathbf{x})$ and $T_{\#}\nu = \nu_{\mathbf{y}}$, then $\{T(\mathbf{x}^{(k)})\}_{1 \leq k \leq N} \stackrel{\text{i.i.d.}}{\sim} \pi_{\mathbf{y}}(\mathbf{x})$



The Transport Map framework

- ▶ Main principle of the **Transport Map** (TM) framework: find a map $T : \mathcal{D}_X \longrightarrow \mathcal{D}_X$ which **pushes forward** the prior measure to the posterior one, *i.e.*: (El Moselhy and Marzouk, 2012)

$$T_{\#}\nu = \nu_y$$

- ▶ The map T is found by minimizing the **Kullback-Leibler Divergence** (KLD) between $T_{\#}\nu$ and ν_y :

$$T = \operatorname{argmin}_{\tilde{T} \in \mathcal{T}_{\Delta}} D_{\text{KL}}(\tilde{T}_{\#}\nu || \nu_y) = \operatorname{argmin}_{\tilde{T} \in \mathcal{T}_{\Delta}} \int_{\mathcal{D}_X} \tilde{T}_{\#}\pi(x) \log \left(\frac{\tilde{T}_{\#}\pi(x)}{\pi_y(x)} \right) dx$$

- ▶ The map T is sought in the set $\mathcal{T}_{\Delta}$ of **triangular monotone** $\mathcal{C}^1$ -diffeomorphisms $\rightarrow$ choice motivated by theoretical results from **Optimal Transport** theory (Villani, 2009)

Computation of transport maps

- ▶ The KLD minimization problem can be reformulated into a **functional minimization** problem:

$$T = \operatorname{argmin}_{\tilde{T} \in \mathcal{T}_\Delta} \mathcal{J}(\tilde{T})$$

where $\mathcal{J} : \mathcal{T}_\Delta \rightarrow \mathbb{R}$ is a functional **which does not depend on the evidence C**

- ▶ Parametrization of maps of $\mathcal{T}_\Delta$ based on **polynomial chaos expansions**:

$$\hat{T}_k(\mathbf{x}_{1:k}; \mathbf{a}) = \sum_{\alpha \in \mathcal{A}_c} a_{c,\alpha} \phi_{c,\alpha}(\mathbf{x}_{1:k-1}) + \int_0^{x_k} \left(\sum_{\alpha \in \mathcal{A}_e} a_{e,\alpha} \phi_{\alpha,e}(\mathbf{x}_{1:k-1}, \xi) \right)^2 d\xi$$

where $(\phi_{c,\alpha})_{\alpha \in \mathcal{A}_c}$, $(\phi_{e,\alpha})_{\alpha \in \mathcal{A}_e}$ are orthogonal polynomials (Marzouk et al., 2016)

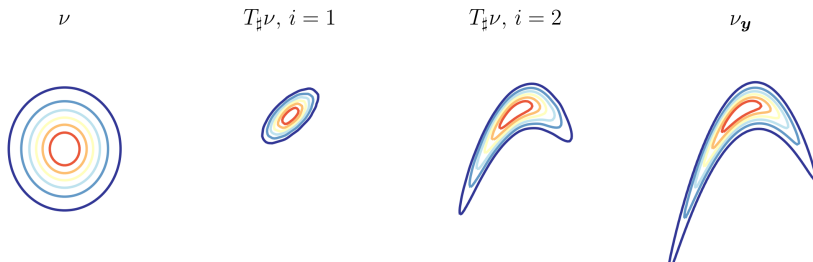
- ▶ The map **parameters** $\mathbf{a} \in \mathbb{R}^P$ are found by solving the following discretized optimization problem:

$$\mathbf{a} = \operatorname{argmin}_{\tilde{\mathbf{a}} \in \mathbb{R}^P} \mathcal{J}(\hat{T}(\cdot; \tilde{\mathbf{a}}))$$

Iterative transport map construction

- ▶ Proposed algorithm: iterative construction of a **composite transport map**
- ▶ The total polynomial degree of the components of T is gradually increased while minimizing the following KLD error estimate:

$$D_{\text{KL}}(T_{\#}\nu || \nu_{\mathbf{y}}) \approx \varepsilon(T) = \frac{1}{2} \text{Var} [\log \pi(\mathbf{X}) - \log T^{\#} \tilde{\pi}_{\mathbf{y}}(\mathbf{X})]$$



Iterative transport map construction

- Proposed algorithm: iterative construction of a **composite transport map**

Algorithm: iterative TM construction

Inputs: error threshold $\varepsilon_0 > 0$, parametrized map space $\mathcal{T}_{\Delta}^h$

1. Initialize $T = \text{Id}$, $\varepsilon(T) = +\infty$, $p = 0$ and $i = 0$
2. While $\varepsilon(T) \geq \varepsilon_0$:
 - 2.1 Set $i = i + 1$ ▷ increase counter
 - 2.2 Set $p = p + 1$ ▷ increase map order
 - 2.3 Compute $T^{(i)} = \underset{\tilde{T} \in \mathcal{T}_{\Delta}^h}{\operatorname{argmin}} D_{\text{KL}}(\tilde{T}_{\#}\nu || T^{\#}\nu_{\mathbf{y}})$ ▷ map at step i
 - 2.4 Set $T = T \circ T^{(i)}$ ▷ update map
 - 2.5 Compute $\varepsilon(T)$ ▷ compute KLD error estimate

Outputs: composite map $T = T^{(1)} \circ \dots \circ T^{(r)}$ (with $T_{\#}\nu \approx \nu_{\mathbf{y}}$)

Polynomial chaos surrogates for accelerated TM inference

- ▶ The KLD minimization problem may involve a large amount of likelihood evaluations ($\mathcal{O}(10^{5-6})$)
- ▶ Use of **polynomial chaos expansions** to approximate the forward model $\mathcal{M}$:

$$\mathcal{M}(\mathbf{X}) \approx \hat{\mathcal{M}}(\mathbf{X}) = \sum_{\alpha \in \mathcal{A}} c_{\alpha} \psi_{\alpha}(\mathbf{X})$$

(Ghanem and Spanos, 1991)

- $\mathcal{A} \subset \mathbb{N}^d$: set of multi-indices
 - c_{α} : coefficients (to be computed)
 - ψ_{α} : multivariate polynomials, orthogonal with respect to the measure ν
- ▶ Non-intrusive calculation of coefficients through **sparse regression** (*Hybrid LARS* algorithm)
(Blatman and Sudret, 2011)
- ▶ Explicit expression for derivatives of $\hat{\mathcal{M}}$ (e.g. $\nabla \hat{\mathcal{M}}$) $\Rightarrow$ use of **derivative-based optimization algorithms** to solve the KLD minimization problem (e.g. BFGS)

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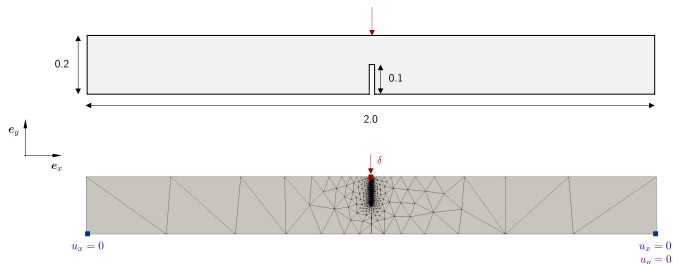
Conclusions

Mechanical inverse problem

- ▶ Finite element model of a concrete beam in a three point bending test
- ▶ Measurements of the force at the top center of the beam at $n = 30$ instants $t_1 < \dots < t_n$

$$\mathbf{y} = (y_1, \dots, y_n)$$

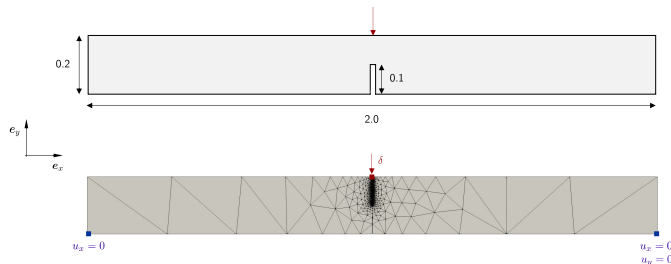
where y_i is the force at the instant t_i (experimental data from (Petersson, 1981))



Mechanical inverse problem

- ▶ Loading: imposed vertical displacement at the top center node of the beam
- ▶ Use of a **Cohesive Zone Model** (CZM) adapted to quasi-brittle materials
- ▶ Finite element calculations performed with the Code_Aster solver (EDF)

(Needleman, 1987)



Mechanical inverse problem

- ▶ 3 uncertain input parameters: Young's modulus, tensile strength, energy release rate
- ▶ **Prior distribution**: mutually independent lognormal marginals

Parameter	Notation	Unit	Distribution	Mean	Coefficient of Variation
Young's modulus	E_c	GPa	Lognormal	35	0.15
Energy release rate	G_c	N.m ⁻¹		130	0.1
Tensile strength	σ_c	MPa		3.5	0.1

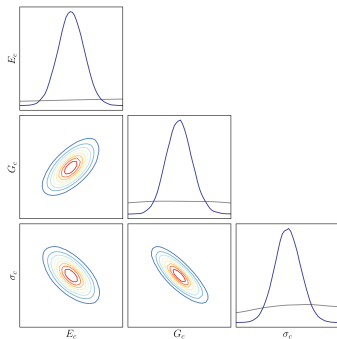
- ▶ **Forward model**: finite element model $\mathcal{M} : \mathcal{D}_X \longrightarrow \mathcal{D}_Y \subset \mathbb{R}^n$ (parameters→force mapping)
- ▶ **Observation model**: additive Gaussian model with known noise variance σ_{ε}^2 :

$$\mathbf{Y} | \mathbf{X} = \mathbf{x} \sim \mathcal{N}(\mathcal{M}(\mathbf{x}), \sigma_{\varepsilon}^2 \mathbf{I})$$

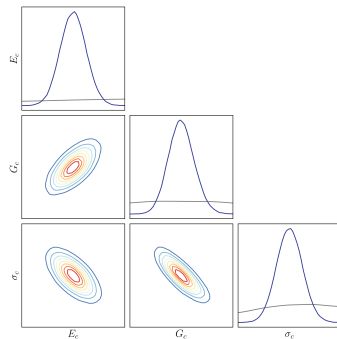
- ▶ **Surrogate modeling**: construction of a prior-based PCE through LARS, with a LHS experimental design of size $K = 500$ (total cost ≈ 2 h)

Bayesian inference - Posterior distribution

- ▶ Application of the iterative TM procedure by replacing the FE model by its PCE surrogate
- ▶ Comparison with MCMC results (*AIES* algorithm, (Goodman and Weare, 2010))



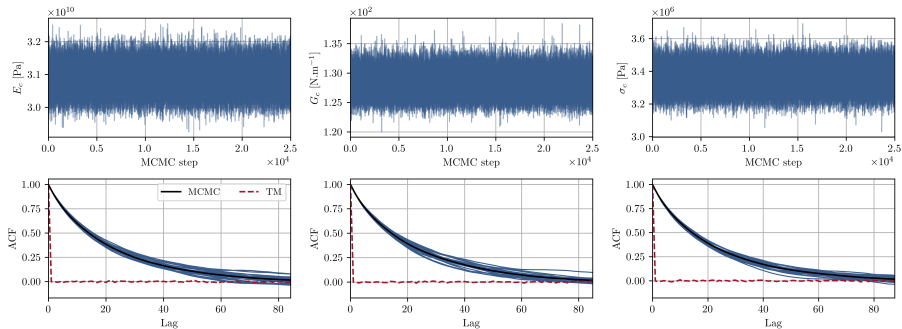
Posterior 1D and 2D marginals - TM



Posterior 1D and 2D marginals - MCMC

Bayesian inference - Posterior distribution

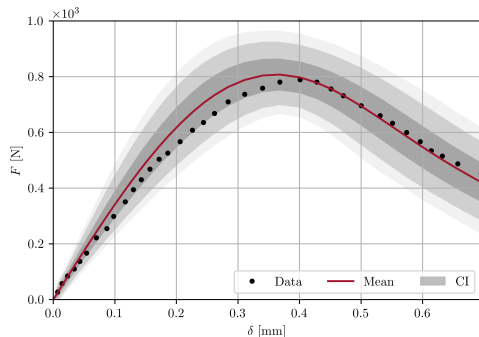
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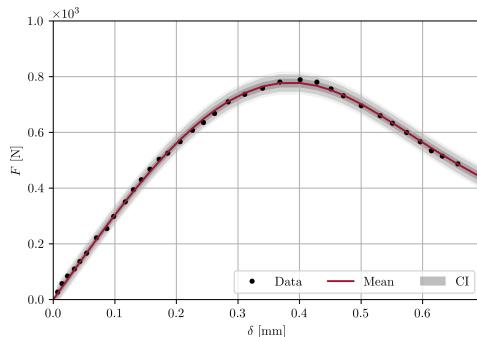
Trace plots and auto-correlation functions - MCMC vs. TM

Bayesian inference - Posterior model predictions

- ▶ Generation of an **arbitrary** number of i.i.d. posterior samples with transport maps
- ▶ Propagation of the posterior samples through the (surrogate) forward model:



Prior force-displacement predictions



Posterior force-displacement predictions

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► Conclusions:

- Transport Maps: *sampling free* approach, avoids the use of MCMC algorithms
- Enables to generate an *arbitrary* number of *i.i.d.* posterior samples
- Use of *PCE surrogates*: ensures a reasonable numerical cost

► Perspectives:

- Construction of *adaptive sparse* transport maps (Baptista et al., 2020)
- Construction of PCE surrogates including derivative information (accuracy on $\mathcal{M} \Rightarrow$ accuracy on $\nabla \mathcal{M}$?)
- Drawbacks of *prior-based* PCE surrogates $\Rightarrow$ use of *adaptive* surrogates, which may be refined throughout TM-based inference

(Li and Marzouk, 2014; Lu et al., 2015)

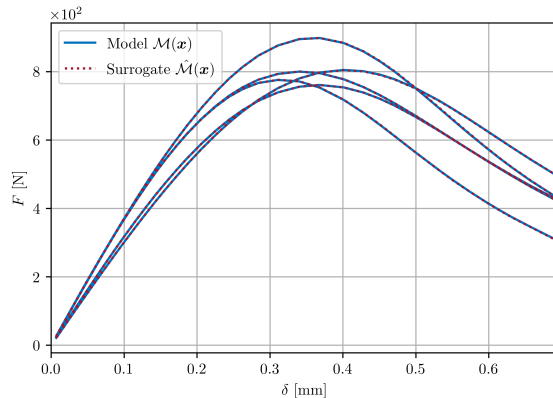
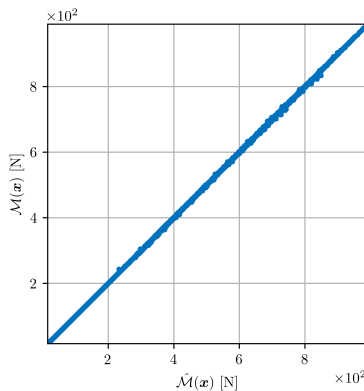
Thank you for your attention !

Back-up slides

PCE surrogate modeling
Prior-based surrogates

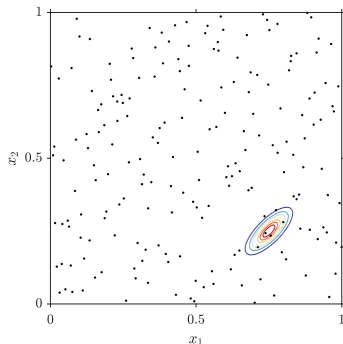
PCE surrogate modeling

- Construction of a PCE surrogate with a LHS experimental design of size $K = 500$, and a maximal total order of 3



Prior-based surrogates

- ▶ Inference with **prior-based** surrogates may fail in the case of concentrated posteriors and/or highly nonlinear forward models (Li and Marzouk, 2014; Lu et al., 2015)
- ▶ The accuracy of prior-based surrogates may not be ensured near informative zones, even in the case of a satisfactory **global** error measure



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